

Exp-1

Comparative study of low-code development versus traditional application development approaches.

Aim:

To compare low-code development and traditional application development approaches based on factors such as development speed, cost, flexibility, scalability, security, maintenance, and suitability for different types of applications, in order to identify their advantages, limitations, and appropriate use cases.

Introduction to Low Code Approach:

The traditional approach to application development is the conventional method of creating software by writing code using programming languages such as Java, Python, C#, or C++. It follows a systematic Software Development Life Cycle (SDLC), which includes planning, designing, coding, testing, deployment, and maintenance. This approach is mainly carried out by skilled software developers.

The traditional approach provides complete control over the application's design, functionality, and performance. It allows developers to build highly customized, secure, and scalable applications based on specific business requirements. This method is commonly used for developing enterprise software, banking systems, healthcare applications, and other complex projects.

Although traditional development takes more time and requires experienced programmers, it is the preferred choice for applications that need high performance, advanced features, and long-term reliability.

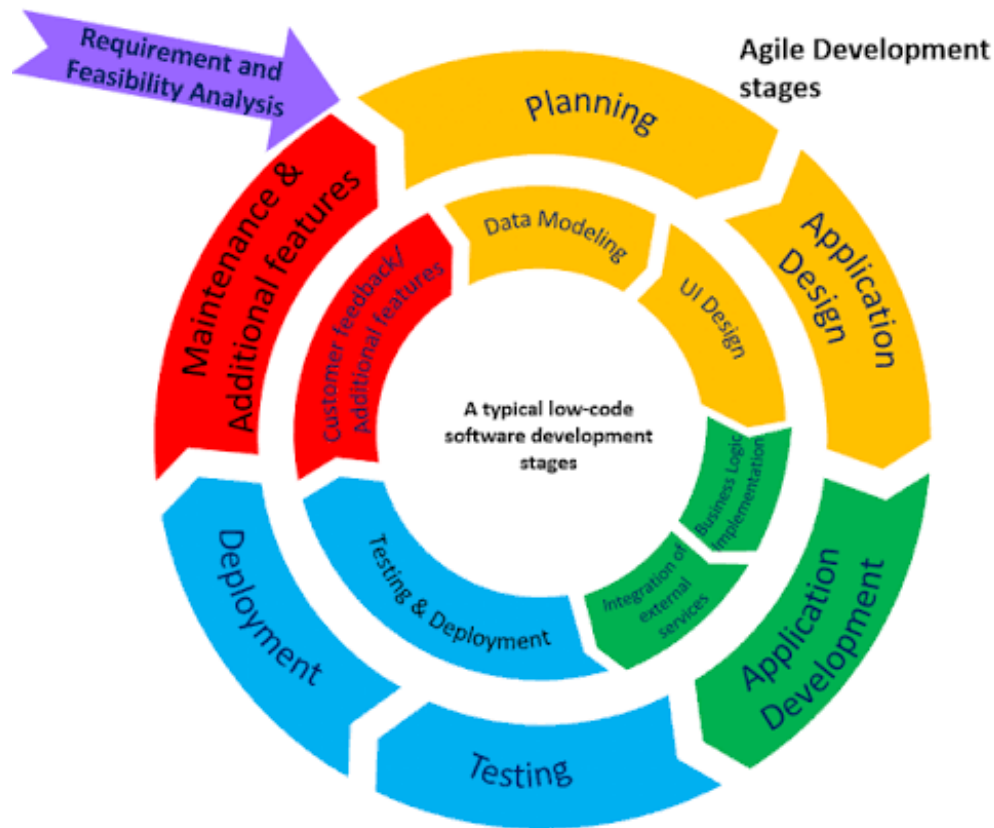
Introduction to Low Code Approach:

The low-code approach is a modern method of application development that allows users to build software with minimal coding. It uses visual tools such as drag-and-drop interfaces, pre-built templates, and reusable components to simplify the development process. This enables both professional developers and non-technical users to create applications quickly.

Low-code platforms help reduce development time and cost by automating many coding tasks. They are widely used to develop business applications, automate workflows, and create web and mobile applications. Organizations use low-code development to speed up digital transformation and deliver software faster.

Although the low-code approach is easy to use and efficient, it may have limitations in customization and performance for highly complex applications. It is best suited for simple to moderately complex applications where rapid development and quick deployment are important.

Diagram for Stages in Traditional And Low Code:



Compare Traditional And Low Code:

Feature	Traditional Application Development	Low-Code Development
Definition	Applications are developed by writing code manually using programming languages.	Applications are developed using visual tools with minimal manual coding.
Development Time	Longer development time.	Faster development and deployment.
Scalability	Highly scalable for large and complex enterprise applications.	Suitable for small to medium-sized applications; scalability depends on the platform.
Performance	Optimized for high performance and complex workloads.	Good for standard business applications but may have performance limitations for highly complex systems.
Security	Developers have full control over security implementation.	Security is largely managed by the platform with less direct control.
Examples	Applications built using Java, Python, C#, PHP, or C++.	Applications built using Microsoft Power Apps, OutSystems, Mendix, Zoho Creator, or Appian.

